

Exercise C1

Q1. If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction -

- (A) $A^n = nA - (n-1)I$ (B) $A^n = 2^{n-1}A - (n-1)I$ (C) $A^n = nA + (n-1)I$ (D) $A^n = 2^{n-1}A + (n-1)I$

Ans. A

Sol. Given, $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

$$\text{Now, } A^2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}; A^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}; A^3 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

Similarly, $A^n = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$ can be verified by induction.

Now, go option by option

$$(b) \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix} \neq \begin{bmatrix} n & 0 \\ n & n \end{bmatrix} + \begin{bmatrix} n-1 & 0 \\ 0 & n-1 \end{bmatrix}$$

$$(d) nA - (n-1)I = \begin{bmatrix} n & 0 \\ n & n \end{bmatrix} + \begin{bmatrix} n-1 & 0 \\ 0 & n-1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$$

Hence, option (d) is correct.

Q2. If $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$ and $x = P^T Q^{2005} P$, then x is equal to

- (A) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 4 + 2005\sqrt{3} & 6015 \\ 2005 & 4 - 2005\sqrt{3} \end{bmatrix}$ (C) $\frac{1}{4} \begin{bmatrix} 2 + \sqrt{3} & 1 \\ -1 & 2 - \sqrt{3} \end{bmatrix}$
 (D) $\frac{1}{4} \begin{bmatrix} 2005 & 2 - \sqrt{3} \\ 2 + \sqrt{3} & 2005 \end{bmatrix}$

Ans. A

Sol. $PP^T = I$

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \text{ \& so on } Q = PAP^T$$

$$Q^2 = (PA^2P^T)(PAP^T) = PA^2P^T$$

$$Q^{2005} = P A^{2005} P^T$$

$$x = P^T (P A^{2005} P^T) P \Rightarrow x = A^{2005} = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$$

Q3.

Let $A = \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix}$ and suppose that $\det(A) = 2$ then the $\det(B)$ equals, where $B = \begin{bmatrix} 4x & 2a & -p \\ 4y & 2b & -q \\ 4z & 2c & -r \end{bmatrix}$

(A) $\det(B) = -2$ (B) $\det(B) = -8$ (C) $\det(B) = -16$ (D) $\det(B) = 8$

Ans. C

Sol. $\det(B) = \begin{vmatrix} 4x & 2a & -p \\ 4y & 2b & -q \\ 4z & 2c & -r \end{vmatrix} = (4)(2)(-1) \begin{vmatrix} x & a & p \\ y & b & q \\ z & c & r \end{vmatrix} = -8 \begin{vmatrix} x & y & z \\ a & b & c \\ p & q & r \end{vmatrix} = -8 \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$
 $= -8 \times 2 = -16$ Ans.

Q4. If A is a non singular matrix satisfying $A = AB - BA$, then which one of the following holds true

(A) $\det. B = 0$ (B) $B = 0$ (C) $\det. A = 1$ (D) $\det. (B + I) = \det. (B - I)$

Ans. D

Sol. A is non singular $\det A \neq 0$

Given $AB - BA = A$ hence $AB = A + BA = A(I + B)$

$$\det. A \cdot \det. B = \det. A \cdot \det. (I + B) \quad (\det A \neq 0)$$

again $AB - A = BA$

$$A(B - I) = BA$$

$$(\det. A) \cdot \det. (B - I) = \det. B \cdot \det. A \Rightarrow \det. (B - I) = \det. (B) \dots\dots\dots(2)$$

from (1) and (2)

$$\det. (B - I) = \det. (B + I) \text{ Ans.}$$

Q5.

$A = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$ & $B = \begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix}$ then $B^T A^T$ is

- (A) a null matrix (B) an identity matrix (C) scalar, but not an identity, matrix
 (D) such that $\text{tr}(B^T A^T) = 4$

Ans. B

Sol. Compute $(AB)^T$ which is an identity matrix and $(AB)^T = B^T A^T$

Q6. If $A = \begin{bmatrix} \cos^2 \alpha & \sin \alpha \cos \alpha \\ \sin \alpha \cos \alpha & \sin^2 \alpha \end{bmatrix}$; $B = \begin{bmatrix} \cos^2 \beta & \sin \beta \cos \beta \\ \sin \beta \cos \beta & \sin^2 \beta \end{bmatrix}$

are such that, AB is a null matrix, then which of the following should necessarily be an odd integral multiple of $\frac{\pi}{2}$.

- (A) α (B) β (C) $\alpha - \beta$ (D) $\alpha + \beta$

Ans. C

Sol. $AB = \begin{bmatrix} \cos^2 \alpha & \sin \alpha \cos \alpha \\ \sin \alpha \cos \alpha & \sin^2 \alpha \end{bmatrix} \begin{bmatrix} \cos^2 \beta & \sin \beta \cos \beta \\ \sin \beta \cos \beta & \sin^2 \beta \end{bmatrix}$

$$= \begin{pmatrix} \cos^2 \alpha \cos^2 \beta + \sin \alpha \cos \alpha \sin \beta \cos \beta & \cos^2 \alpha \sin \beta \cos \beta + \sin \alpha \cos \alpha \sin^2 \beta \\ \cos^2 \beta \sin \alpha \cos \alpha + \sin^2 \alpha \sin \beta \cos \beta & \sin \alpha \cos \alpha \sin \beta \cos \beta + \sin^2 \alpha \sin^2 \beta \end{pmatrix}$$

$$= \begin{pmatrix} \cos \alpha \cos \beta \cos(\alpha - \beta) & \cos \alpha \sin \beta \cos(\alpha - \beta) \\ \sin \alpha \cos \beta \cos(\alpha - \beta) & \sin \alpha \sin \beta \cos(\alpha - \beta) \end{pmatrix}$$

$\Rightarrow$ must be an odd integral multiple of $\pi/2 \Rightarrow$ (C)

Q7. Let A, B, C, D be (not necessarily square) real matrices such that $A^T = BCD$; $B^T = CDA$; $C^T = DAB$ and $D^T = ABC$ for the matrix $S = ABCD$, consider the two statements.

I $S^3 = S$

II $S^2 = S^4$

- (A) II is true but not I (B) I is true but not II (C) both I and II are true
(D) both I and II are false.

Ans. A

Sol. $S = ABCD = A(BCD) = AA^T$ (1)

$$S^3 = (ABCD)(ABCD)(ABCD)$$

$$= (ABC)(DAB)(CDA)(BCD)$$

$$= D^T C^T B^T A^T$$

$$= (BCD)^T A^T = AA^T \quad \text{.....(2)}$$

from (1) and (2) $S = S^3 \Rightarrow$ I is correct

multiply both sides by S

$$S^2 = S^4 \Rightarrow \text{II is correct} \Rightarrow \text{Both I and II are true Ans.}$$

Q8. There are two possible values of A in the solution of the matrix equation

$$\begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1} \begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix} \text{ where A, B, C, D, E, F are real numbers. The absolute value of the difference of these two solutions, is}$$

- (A) $\frac{8}{3}$ (B) $\frac{11}{3}$ (C) $\frac{1}{3}$ (D) $\frac{19}{3}$

Ans. D

Sol. If $P = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{adj } P = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$$\text{also } A^{-1} = \frac{\text{adj. } A}{\det. A}$$

$$\therefore \begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix} = \frac{\begin{bmatrix} A & 5 \\ 4 & 2A+1 \end{bmatrix}}{2A^2 + A - 20}$$

$$\therefore \frac{1}{2A^2 + A - 20} \begin{bmatrix} A & 5 \\ 4 & 2A+1 \end{bmatrix} \begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$$

$$\text{The gives, } \frac{A^2 - 5A - 10}{2A^2 + A - 20} = 14 \Rightarrow A = 3 \text{ or } \frac{10}{3}$$

$$\therefore 3 + \frac{10}{3} = \frac{19}{3} \text{ Ans.}$$

Q9. Number of real values of λ for which the matrix $A = \begin{bmatrix} \lambda - 1 & \lambda & \lambda + 1 \\ 2 & -1 & 3 \\ \lambda + 3 & \lambda - 2 & \lambda + 7 \end{bmatrix}$ has no inverse

(A) 0 (B) 1 (C) 2 (D) infinite

Ans. D

Sol. $|A| = 0$ for all $\lambda \in \mathbb{R} \Rightarrow A$ is singular. Hence can not be found for any value of $\lambda \in \mathbb{R} \Rightarrow$ (D)

$$\text{Use } R_2 \rightarrow R_2 - \frac{1}{2} (R_3 - R_1)$$

Q10. Let $A = \begin{bmatrix} l & m & n \\ p & q & r \\ 1 & 1 & 1 \end{bmatrix}$ and $B = A^2$.

If $(l - m)^2 + (p - q)^2 = 9$, $(m - n)^2 + (q - r)^2 = 16$, $(n - l)^2 + (r - p)^2 = 25$, then the value of $\det. B$ equals

(A) 36 (B) 100 (C) 144 (D) 169

Ans. C

Sol. $\det. A$ is twice the area of the triangle with vertices (l, p) , (m, q) , (n, r) with sides 3, 4, 5.

$$\Delta^2 = s(s - a)(s - b)(s - c)$$

$$\Delta^2 = 6(6 - 3)(6 - 4)(6 - 5)$$

$$\Delta^2 = 36 \Rightarrow \Delta = 6$$

$$\text{Now } \det A = 2\Delta = 12 \Rightarrow \det. B = (\det A)^2 = 4\Delta^2 = 144 \text{ Ans.}$$

Q11. In a square matrix A of order 3 the elements, a_{ii} 's are the sum of the roots of the equation

$x^2 - (a + b)x + ab = 0$; $a_{ii}, i + 1$'s are the product of the roots, $a_{ii}, i - 1$'s are all unity and the rest of the elements are all zero. The value of the $\det. (A)$ is equal to

(A) 0 (B) $(a + b)^3$ (C) $a^3 - b^3$ (D) $(a^2 + b^2)(a + b)$

Ans. D

Sol. Given $a_{11} = a_{22} = a_{33} = a + b$

$$a_{12} = a_{23} = ab; a_{21} = 1 = a_{32}, \text{ all others are zero.}$$

$$\text{Det (A)} = \begin{vmatrix} a+b & ab & 0 \\ 1 & a+b & ab \\ 0 & 1 & a+b \end{vmatrix} = (a^2 + b^2)(a + b)$$

Q12. Let $A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -\alpha & 14\alpha & 7\alpha \\ 0 & 1 & 0 \\ \alpha & -4\alpha & -2\alpha \end{bmatrix}$. If $AB = I$, where I is an identity matrix

of order 3 then trace B has value equal to

(A) 0 (B) $\frac{2}{5}$ (C) $\frac{1}{5}$ (D) 5

Ans. B

Sol. We have $AB = I$, so

$$\begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} -\alpha & 14\alpha & 7\alpha \\ 0 & 1 & 0 \\ \alpha & -4\alpha & -2\alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5\alpha & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10\alpha - 2 & -5\alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ (Given)}$$

$$\Rightarrow 5\alpha = 1 \Rightarrow \alpha = \frac{1}{5}$$

$$\text{Now, trace (B)} = (-\alpha) + 1 + (-2\alpha) = 1 - 3\alpha = 1 - 3\left(\frac{1}{5}\right) = 1 - \frac{3}{5} = \frac{2}{5} \text{ Ans.}$$

Q13. Let D_k is the $k \times k$ matrix with 0's in the main diagonal, unity as the element of 1st row and column $(f(k))^{th}$ and k for all other entries. If $f(x) = x - \{x\}$ where $\{x\}$ denotes the fractional part function then the value of $\det. (D_2) + \det. (D_3)$ equals

(A) 32 (B) 34 (C) 36 (D) None

Ans. B

Sol. $\text{Det. (D}_2) = \begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix}$ $(f(2) = 2 \Rightarrow 1^{\text{st}} \text{ and } 2^{\text{nd}} \text{ column} = 1)$

|||ly $\text{Det. (D}_3) = \begin{vmatrix} 0 & 3 & 1 \\ 3 & 0 & 3 \\ 3 & 3 & 0 \end{vmatrix}$ $(f(3) = 3 \Rightarrow a_{13} = 1)$

$$\text{Det. (D}_2) = -2; \text{Det. (D}_3) = 36$$

$$\Rightarrow \text{Det. (D}_2) + \text{Det. (D}_3) = 34 \text{ Ans.}$$

Q14. Consider a matrix $A(\theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}$ then

- (A) $A(\theta)$ is symmetric (B) $A(\theta)$ is skew symmetric (C) $A^{-1}(\theta) = A(\pi - \theta)$
 (D) $A^2(\theta) = A\left(\frac{\pi}{2} - 2\theta\right)$

Ans. C

Sol. As $A(\theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}$

$A(\theta)$ is certainly neither symmetric nor skew symmetric

Further $A(\pi - \theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ \cos \theta & \sin \theta \end{bmatrix}$

and $A(\theta) \cdot A(\pi - \theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix} \begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \Rightarrow A^{-1}(\theta) = A(\pi - \theta)$

$A^2(\theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix} \begin{bmatrix} -\cos 2\theta & \sin 2\theta \\ -\sin 2\theta & -\cos 2\theta \end{bmatrix} \neq A\left(\frac{\pi}{2} - 2\theta\right)$

Q15. Let A and B be 3×3 symmetric matrices such that $X = AB + BA$ and $Y = AB - BA$,

then $(XY)^T$ is equal to

[Note : $(XY)^T$ is the transpose of the matrix XY .]

- (A) XY (B) YX (C) $-XY$ (D) $-YX$

Ans. D

Sol. We have $A = A^T$ and $B = B^T$

$$X = AB + BA \Rightarrow X = X^T \text{ and } Y = AB - BA \Rightarrow Y = -Y^T$$

Now, $(XY)^T = Y^T X^T = -YX$ Ans.

Q16. If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 0 \\ -1 & 1 \end{bmatrix}$, then number of distinct values of α for which $A^2 = B$ equals

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol. We have

$A^2 = B$ (Given)

$$\begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ -1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} \alpha^2 & 0 \\ \alpha + 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ -1 & 1 \end{bmatrix}$$

So, $\alpha^2 = 4$ and $\alpha + 1 = -1 \Rightarrow \alpha = \pm 2$ and $\alpha = -2$

$\therefore \alpha = -1$

Hence, only one possible value of $\alpha = -2$ exist. Ans.

Q17. If A is a diagonal matrix of order 3 such that $A^2 = A$, then number of possible matrices A, is ($A \neq O$)

(A) 2 (B) 3 (C) 8 (D) 7

Ans. D

Sol. $A^2 =$

$\Rightarrow d_1, d_2, \dots, d_n$ be the diagonal matrix of A, then

$$A^2 = A \Rightarrow d_1^2 = d_1, d_2^2 = d_2, \dots, d_n^2 = d_n$$

Now, $d_i^2 = d_i \Rightarrow$ either $d_i = 0$ or $d_i = 1$

Number of ways in which diagonal elements can be filled up is 2^3 , but all diagonal elements cannot be zero.

$\therefore$ Number of required matrices is $2^3 - 1 = 7$ Ans.

Q18. If $w \neq 1$ is the complex cube root of unity and matrix $H = \begin{bmatrix} w & 0 \\ 0 & w \end{bmatrix}$, then H^{70} is equal to :

(A) H (B) 0 (C) $-H$ (D) H_2

Ans. A

Sol. Since H is a diagonal matrix. We know that product of two diagonal matrix is always a diagonal matrix.

$$\begin{aligned} \text{So } H^{70} &= \begin{bmatrix} w & 0 \\ 0 & w \end{bmatrix} \begin{bmatrix} w & 0 \\ 0 & w \end{bmatrix} \cdots \begin{bmatrix} w & 0 \\ 0 & w \end{bmatrix} \text{ 70 times} \\ &= \begin{bmatrix} w^{70} & 0 \\ 0 & w^{70} \end{bmatrix} = \begin{bmatrix} w & 0 \\ 0 & w \end{bmatrix} = H \end{aligned}$$

Q19. Let $w \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form

$$\begin{bmatrix} 1 & a & b \\ w & 1 & c \\ w^2 & w & 1 \end{bmatrix}, \text{ where each of a, b and c is either w or } w^2. \text{ Then the number of distinct matrices in}$$

the set S is :

(A) 2 (B) 6 (C) 4 (D) 8

Ans. A

Sol. $\begin{vmatrix} 1 & a & b \\ w & 1 & c \\ w^2 & w & 1 \end{vmatrix}$

$$=1 - cw - a(w - w^2c) = (1 - cw) - aw(1 - cw) = (1 - cw)(1 - aw)$$

for non singular matrix

$$c \neq \frac{1}{w} \text{ and } a \neq \frac{1}{w}$$

$$\Rightarrow c \neq w^2, \quad a \neq w^2$$

$$\Rightarrow a \text{ and } c \text{ must be } w \text{ and } b \text{ can be } w \text{ or } w^2$$

$$\therefore \text{ total matrices} = 2$$

Q20. If A and B are different matrices satisfying $A^3 = B^3$ and $A^2B = B^2A$, then

(A) $\det(A^2 + B^2)$ must be zero. (B) $\det(A - B)$ must be zero.

(C) $\det(A^2 + B^2)$ as well as $\det(A - B)$ must be zero.

(D) At least one of $\det(A^2 + B^2)$ or $\det(A - B)$ must be zero.

Ans. D, A

Sol. $A^3 = B^3$ (1)

and $A^2B = B^2A$ (2)

(1) - (2) gives, $A^3 - A^2B = B^3 - B^2A$

$$A^2(A - B) = -B^2(A - B) \Rightarrow (A^2 + B^2)(A - B) = 0$$

$$\det(A^2 + B^2)(A - B) = 0$$

$$\det(A^2 + B^2) \cdot \det(A - B) = 0 \Rightarrow (D)$$

Q21. Given the matrices A and B as $A = \begin{bmatrix} 1 & -1 \\ 4 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 4 & -2 \end{bmatrix}$.

The two matrices X and Y are such that $XA = B$ and $AY = B$ then which of the following hold(s) true?

(A) $X = \frac{1}{3} \begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix}$ (B) $Y = \frac{1}{3} \begin{bmatrix} 3 & 0 \\ 4 & 0 \end{bmatrix}$ (C) $\det. X = \det. Y$

(D) $3(X + Y) = \begin{bmatrix} 4 & -1 \\ 4 & 2 \end{bmatrix}$

Ans. D, C

Sol. Note that A is non singular but B is singular hence only A^{-1} exists

Now $XA = B \Rightarrow X = BA^{-1}$ (1)

and $AY = B \Rightarrow Y = A^{-1}B$ (2)

Also $A^{-1} = \frac{1}{3} \begin{bmatrix} -1 & 1 \\ -4 & 1 \end{bmatrix}$

now verify

Q22. If there are three square matrix A, B, C of same order satisfying the equation $A^2 = A^{-1}$ and let $B = A^{2^n}$ & $C = A^{2^{(n-2)}}$ then which of the following statements are true?

(A) $\det. (B - C) = 0$ (B) $(B + C)(B - C) = 0$ (C) B must be equal to C (D) none

Ans. A, B, C

Sol. $B = A^{2^n} = A^{2 \cdot 2^{n-1}} = (A^2)^{2^{n-1}} = (A^{-1})^{2^{n-1}} = (A^{2^{n-1}})^{-1} = (A^{2 \cdot 2^{n-2}})^{-1} = ((A^2)^{2^{n-2}})^{-1}$

$$((A^{-1})^{-1})^{2^{(n-2)}} = A^{2^{(n-2)}}$$

so $B = C$

$$(B - C) = \text{null} \quad (B + C)(B - C) = 0 \quad \Rightarrow \quad \det. (B - C) = 0$$

$\Rightarrow$ A, B, C are answers